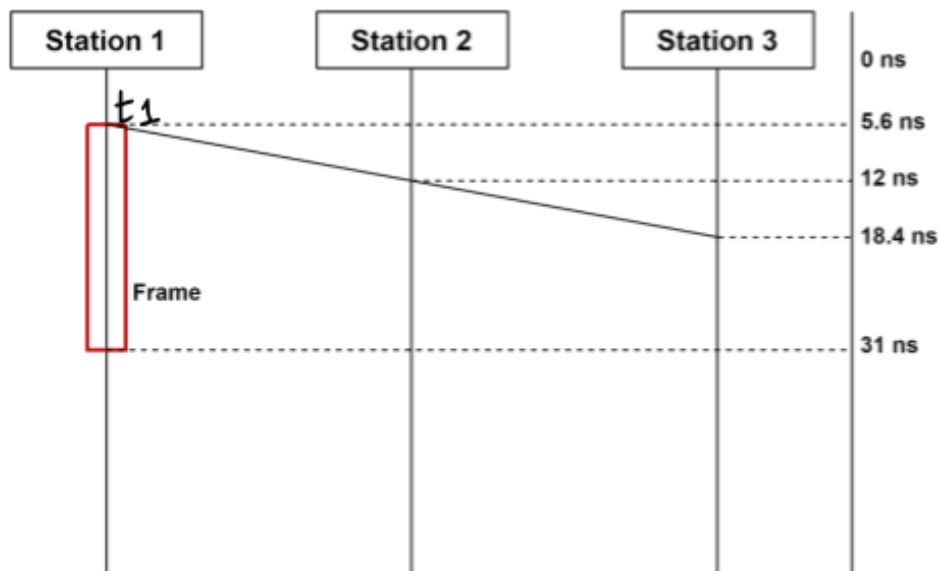


CSE320 (Data Comm) – Practice Sheet on Chapter 12

Q1.

Suppose, all the stations are transmitting following CSMA/CD protocol.



Station 1 and Station 3 are at opposite ends of the common multipoint link that all the stations share. The bandwidth of the link is 10 Gbps.

- Find out the transmission time and maximum propagation time from the given diagram.
- If collision detection methods are introduced, what should be the minimum size of the frame in bytes? Does the frame in the given diagram meet the minimum size requirement?

Q2.

A common multipoint link is being used by three stations. Station 1 is to the furthest left and Station 3 is to the furthest right, with the remaining stations in between.

- Station 1 starts to send a frame to Station 3 at time $t = 4$ ms.
- The first bit of the frame reaches Stations 2 and 3 at times $t = 13.4$ ms and 17.1 ms respectively.
- Station 1 is supposed to send the last bit of the frame at time $t = 21.4$ ms.

- Sketch a space and time representation of the channel if it is using CSMA/CD.

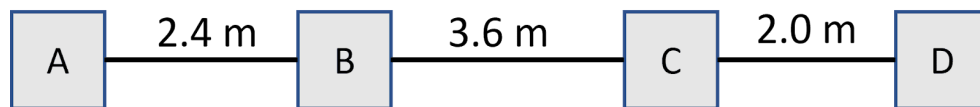
- ii. What is the Transmission Time and Maximum Propagation Time from the given information?
- iii. What should be the Minimum Frame Size and does it meet the requirements given above if the bandwidth is 2 Gbps?

Q3.

For $K=3$, what is the size of the contention window in CSMA/CA? During the waiting time of the contention window slots, if the station finds the line busy, does the station halt or restart the contention window?

Q4.

Four devices are sharing a single communication channel using the CSMA/CD protocol as shown in the figure below. Propagation speed of the medium is 2×10^6 m/s.



- Suppose, device A wants to send a frame to another device. If the channel bandwidth is 2 Mbps (megabits per sec), what will be the minimum frame size required? What if the frame size is smaller than that?
- Suppose, device B starts sending a frame at time $2.0 \mu\text{s}$ (microsecond). Device D starts sending a frame at $4.5 \mu\text{s}$. Find out whether this will lead to a collision.

Q5.

Three devices are sharing a common channel using pure ALOHA protocol. Frame transmission time is $10 \mu\text{s}$ (each frame takes $10 \mu\text{s}$ to transmit).

- Device A generates TWO frames at times $10 \mu\text{s}$ and $40 \mu\text{s}$.
- Device B generates TWO frames at times $20 \mu\text{s}$ and $35 \mu\text{s}$.
- Device C generates ONE frame at time $5 \mu\text{s}$.

Using a figure, find out how many frames will be successfully transmitted.